

Coevolution in the tumor microenvironment

Robert A Weinberg

The progression of carcinomas to high-grade malignancies is accompanied by profound histological changes in the tumor-associated stroma. Although previous studies have suggested that mesenchymal cells of the stroma undergo genetic alterations during this progression, a new study now provides evidence that strongly contradicts this theory of stromal cell coevolution.

Cancer research over the past three decades has focused on the neoplastic cells within tumors. However, in recent years, tumors have come to be understood to function as complex tissues in which numerous recruited host cells also play critical roles. These recruited cells are of mesenchymal origin and, in aggregate, they constitute the tumor-associated stroma. In many tumors, these stromal cells greatly exceed in number the neoplastic cells that drive tumor growth. Some have proposed that stromal cells coevolve with cancer cells and undergo alterations in genotype and phenotype. Now, in a new study on page 650 of this issue, Ian Campbell and colleagues report that their survey of the genomes of isolated stromal cells has convincingly failed to find evidence of widespread genetic changes, calling into question earlier reports of stromal genome alterations¹.

Tumor stromal cells

Among the tumor-associated mesenchymal cells are endothelial cells and pericytes that together form the neovasculature, which supplies tumor cells with nutrients and oxygen and evacuates wastes and carbon dioxide. These blood vessels within tumors are overshadowed by vastly larger numbers of fibroblasts and myofibroblasts. Fibroblasts have key roles in maintaining and supporting normal epithelial tissues, whereas myofibroblasts are rarely seen in normal tissues, but instead participate in tissue reconstruction as components of wound healing and in pathological processes such as fibrosis. The transient presence of myofibroblasts in sites of wound healing contrasts with their increasing numbers as carcinomas progress to malignancy. Indeed, this behavior has provided support for the notion that tumors are

'wounds that do not heal'². The implication is that carcinomas appropriate the normal wound-healing response in order to acquire the types of physiologic support needed for their continued growth.

The development of human carcinomas usually occurs over a time span of decades. As the mesenchymal cells of the tumor stroma are already apparent in tumors at their earliest stages, this suggests that these mesenchymal cell populations reside within tumors for periods of time comparable to those of the neoplastic cell populations that recruited them in the first place. Because the neoplastic cell populations often progressively evolve to higher degrees of malignancy, it becomes plausible that the initially recruited stromal cells coevolve with the cancer cells—that is, that they continually alter their genotype and phenotype in order to accommodate the needs of their ever-changing neoplastic neighbors. Such a theory of stromal coevolution suggests that stromal cells occasionally stumble on more advantageous genotypes and phenotypes and that these acquired changes confer an advantage to the stromal cells, thereby leading to clonal expansions within the stroma. This model also suggests that the fibroblasts and myofibroblasts that are initially recruited into the tumor stroma are not up to the task of supporting advanced tumors and require additional changes to render them more capable of fostering tumor growth.

All things considered, arguments over stromal coevolution hold little weight in comparison with direct measurements of genetic change of stromal cell genomes within actual human tumors. This type of experiment is challenging, because the stromal and epithelial layers of tumors are often closely interwoven (Fig. 1), and careful microdissection is necessary to ensure that stromal cell preparations are not contaminated with carcinoma cells.

Do stromal cells coevolve?

A number of studies have indeed shown often-striking changes in the genomes of

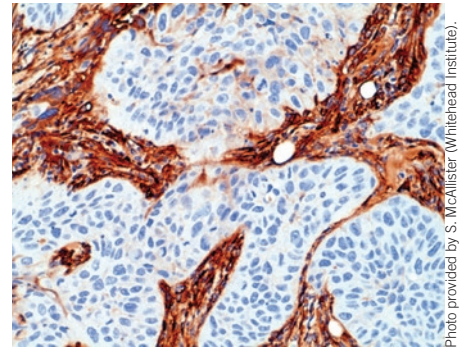


Figure 1 Islands of neoplastic epithelial cells (light blue nuclei) are surrounded by a myofibroblast-rich stroma (dark brown).

tumor-associated stromal cells^{3–10}, including loss of heterozygosity and changes in copy number of certain loci as well as frequent somatic mutations of key tumor suppressor genes, notably *PTEN* and *TP53*. In contrast, Qiu *et al.*¹ did not find genetic alterations in carcinoma-associated fibroblasts from breast and ovarian tumors, even though rampant changes were apparent within the genomes of neighboring epithelial cells with the same carcinomas. The authors profiled isolated tumor and cancer-associated fibroblast preparations for genome-wide copy number and loss of heterozygosity using array analysis for hundreds of thousands of SNPs. In addition, Qiu *et al.*¹ used microsatellite marker analyses to verify that there were no allelic imbalances in their stromal cell preparations at loci that were reported in previous studies to show high frequencies of loss of heterozygosity in ovarian and breast cancer stromal cells. A similar study¹¹ failed to find genetic alterations in stromal cells from ductal breast carcinomas studied by SNP array analysis for tens of thousands of SNPs.

There are, however, potentially important technical differences between these studies. These two reports each used fresh-frozen rather than paraffin-fixed tissues in their analyses. Further, their analyses depended largely on use of SNP arrays rather than, for

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example, nested PCR for genomic analyses. But in the end, independent of the deployed techniques, the relative weights of the conflicting conclusions are hardly equal. The observation of extensive genetic changes in stromal cells might be dismissed as reflective of contamination by nearby carcinoma cells or damage to DNA during the course of sample preparation and analysis. However, the finding of pristine genomes in stromal cells cannot be so readily dismissed as experimental artifact.

In conclusion, although these two papers provide convincing evidence to question the stromal coevolution theory, they are more

likely to raise more questions than to settle this dispute. A fully controlled, side-by-side comparison of the analytical procedures (using different methods of preparing tissue and detecting genomic alterations) will be required to reach an adequately definitive resolution to convince the field to let go of the concept of coevolution of stromal cells. Conclusion of this controversy is surely important to those who wish to understand tumor biology, as the genetic coevolution of stromal cells, if real, would create an entirely new dimension of complexity that would further complicate our already-intricate depictions of tumor pathogenesis.

1. Qiu, W. *et al.* *Nat. Genet.* **40**, 650–655 (2008).
2. Dvorak, H.F. *N. Engl. J. Med.* **315**, 1650–1659 (1986).
3. Lakhani, S.R. *et al.* *J. Pathol.* **189**, 496–503 (1999).
4. Moynfar, F. *et al.* *Cancer Res.* **60**, 2562–2566 (2000).
5. Wernert, N., Locherbach, C., Wellmann, A., Behrens, P. & Hugel, A. *Anticancer Res.* **21**, 2259–2264 (2001).
6. Kurose, K. *et al.* *Nat. Genet.* **32**, 355–357 (2002).
7. Patocs, A. *et al.* *N. Engl. J. Med.* **357**, 2543–2551 (2007).
8. Fukino, K., Shen, L., Patocs, A., Mutter, G.L. & Eng, C. *J. Am. Med. Assoc.* **297**, 2103–2111 (2007).
9. Weber, F. *et al.* *J. Am. Med. Assoc.* **297**, 187–195 (2007).
10. Moynfar, F. *et al.* *Mod. Pathol.* doi:10.1038/modpathol.2008.28 (8 February 2008).
11. Allinen, M. *et al.* *Cancer Cell* **6**, 17–32 (2004).

Salt wasting and blood pressure

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Recessive loss-of-function mutations in genes involved in renal NaCl handling cause rare diseases characterized by salt wasting and reduced blood pressure of variable severity. A new study shows that the carrier state for rare inactivating mutations in three genes involved in NaCl transport in the kidney is associated with a significant blood pressure reduction and a reduced risk of hypertension in the general population.

Every day, an individual's kidneys filter 180 l of plasma and reabsorb more than 99% of the filtered sodium, corresponding to over 1 kg of salt. This massive reabsorption of NaCl results from transport mechanisms operating in the epithelial cells lining specialized tubular segments. Richard Lifton and colleagues have previously provided critical insights into the role these processes have in homeostasis, with studies demonstrating that rare mendelian diseases altering renal salt handling significantly affect blood pressure regulation¹. These diseases include the recessively inherited Bartter's and Gitelman's syndromes, which lead to renal salt wasting potentially associated with life-threatening hypotension and electrolyte disturbances. On page 592 of this issue, Weizhen Ji, Richard Lifton and colleagues² now take this one step further and provide persuasive data suggesting that the carrier state for rare functional mutations in three genes involved in Bartter's and Gitelman's syndromes influences blood pressure regulation in the general population.

Rare salt-losing tubulopathies

The appropriate handling of water and

solutes by the kidney is dependent on the coordinated action of specialized tubular segments. Approximately 25% of the filtered NaCl is reabsorbed by the thick ascending limb (TAL) of the loop of Henle. The process is inhibited by loop diuretics, which are powerful drugs used to treat states of NaCl retention. The distal convoluted tubule (DCT) reabsorbs 5–10% of the filtered NaCl through a pathway that is inhibited by thiazide diuretics, which are widely used as antihypertensive drugs (Fig. 1). The identification of these transport processes was facilitated by the elucidation of rare, recessively inherited tubulopathies including antenatal Bartter's syndrome, associated with massive renal salt wasting and early lethality, and Gitelman's syndrome, showing milder, later-onset and thiazide-like salt loss with hypokalemia and hypomagnesemia³. A decade ago, Lifton and colleagues demonstrated that antenatal Bartter's syndrome results from inactivating mutations in *SLC12A1* or *KCNJ1*, the genes coding for the Na⁺-K⁺-2Cl⁻ cotransporter NKCC2 and the K⁺ channel ROMK in the TAL, respectively, whereas Gitelman's syndrome is associated with loss-of-function mutations in *SLC12A3*, encoding the thiazide-sensitive Na-Cl cotransporter NCCT in the DCT¹. To date, 22 mutations in *SLC12A1*, 35 in *KCNJ1* and 110 in *SLC12A3* have been reported. Most individuals with

Bartter's and Gitelman's syndromes harbor these mutations at the homozygous or compound heterozygous state. Several classes of *SLC12A3* mutants have been identified *in vitro*, potentially accounting for the phenotype variability in Gitelman's syndrome⁴. The identification of two other genes involved in the basolateral transport of Cl⁻ (refs. 5,6) led to a comprehensive classification of the salt-losing tubulopathies originating in the TAL and DCT³.

Weight of the carrier state

Gitelman's and Bartter's syndromes are rare diseases, with prevalence estimated at ~20 per million and ~1 per million, respectively, reflecting their variable severity and age of onset². Accordingly, the prevalence of carriers of *SLC12A3* mutations is ~1% in the general population. Ji *et al.* screened 3,125 subjects from the Framingham Heart Study (FHS) for mutations in *SLC12A3*, *SLC12A1* and *KCNJ1* to test whether single loss-of-function mutations in these genes may affect blood pressure regulation in this population². The choice of the FHS cohort was motivated by the possibility to obtain a standardized long-term assessment of systolic and diastolic blood pressure adjusted for age, sex and antihypertensive treatment. Ji *et al.* identified a total of 138 variants in the coding sequence of 2,492 subjects. By comparing these with known

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